



**Pimpri Chinchwad Education Trust's
Pimpri Chinchwad College of Engineering**

FA-1 Activity

Department: Computer Engineering

Academic Year: 2025-26

Semester: V

Class: T.Y. B.Tech

Div.: D

Date: 11/09/2025

Subject: DMW

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PRN: 123B1B220

Question No.: 14

Problem Statement: OLAP – Hospital Appointments

- Create hierarchy (Year -> Month -> Day) for Appointments
- Roll-up data from **Doctor -> Department -> Hospital** and show the appointment count

Solution:

Hospital Appointment Monitoring System

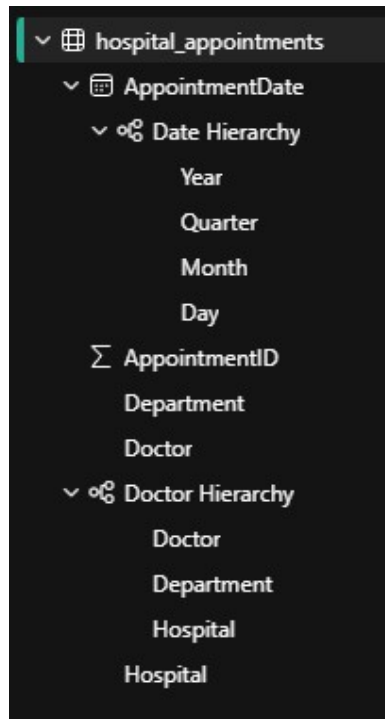
1. Dataset Information

The dataset used for this analysis contains **120 hospital appointment records** for **CityCare Hospital**. It includes:

- 10 Doctors** assigned across **4 Departments** (Cardiology, Neurology, Orthopedics, Pediatrics).
- A single hospital entity (CityCare Hospital).
- Appointment details with **Appointment ID** and **Appointment Date** (supporting Year → Month → Day hierarchy).

This dataset was created to support OLAP-style analysis of hospital appointments.

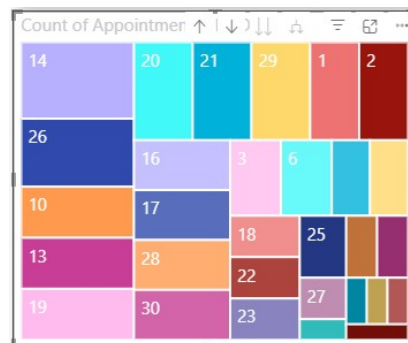
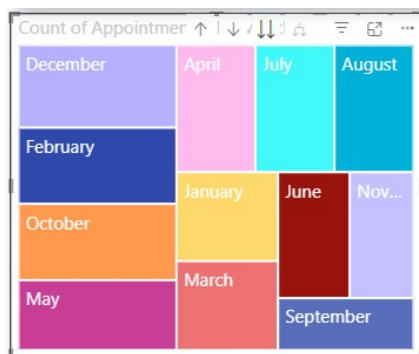
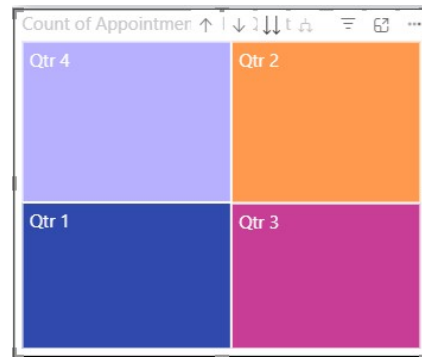
AppointmentID	Doctor	Department	Hospital	AppointmentDate
1	Doctor_4	Neurology	CityCare Hospital	15 February 2024
2	Doctor_4	Neurology	CityCare Hospital	19 February 2024
3	Doctor_3	Cardiology	CityCare Hospital	15 February 2024
4	Doctor_3	Cardiology	CityCare Hospital	29 October 2024
5	Doctor_6	Orthopedics	CityCare Hospital	29 December 2024
6	Doctor_8	Neurology	CityCare Hospital	26 July 2024
7	Doctor_1	Pediatrics	CityCare Hospital	18 December 2024
8	Doctor_1	Pediatrics	CityCare Hospital	18 January 2024
9	Doctor_1	Pediatrics	CityCare Hospital	22 May 2024
10	Doctor_5	Orthopedics	CityCare Hospital	25 April 2024
11	Doctor_2	Pediatrics	CityCare Hospital	18 July 2024



2. Solution for the Problem Statements

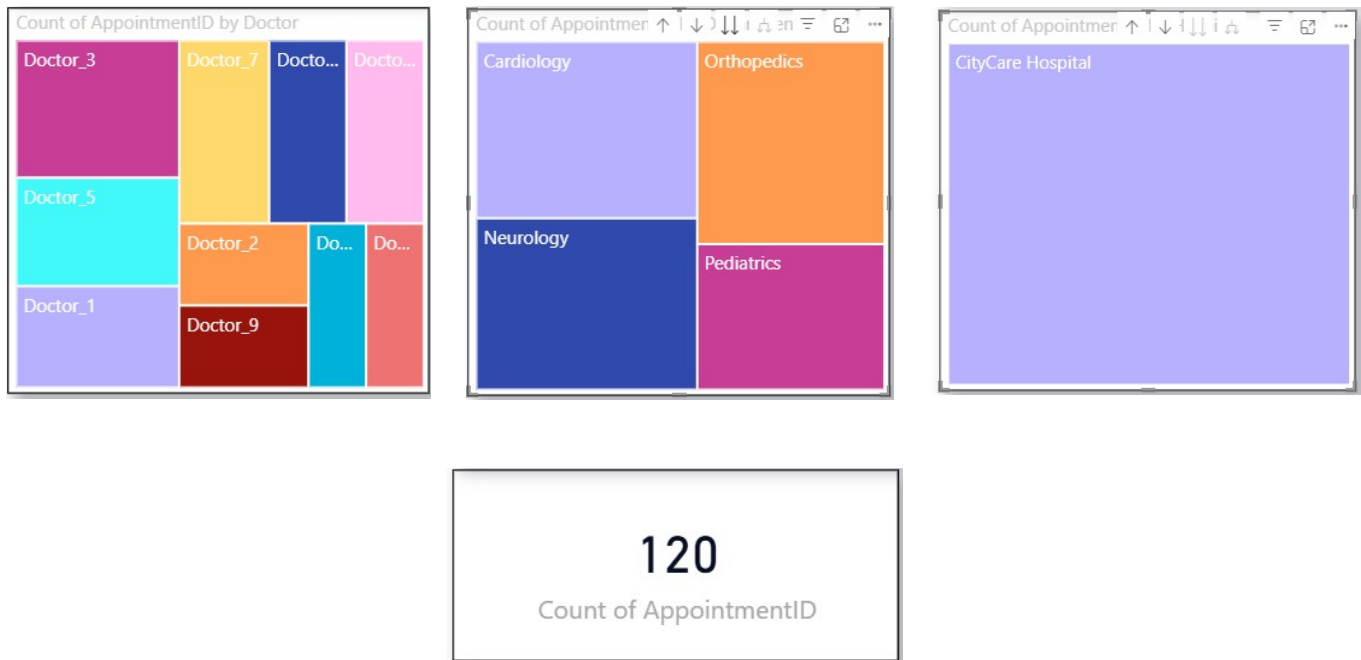
a) Hierarchy (Year → Month → Day)

- Implemented a **Tree Graph** showing the **date hierarchy** for appointments, allowing drill-down from year → month → day.
- Enables time-based trend analysis of hospital appointments.



b) Roll-Up (Doctor → Department → Hospital)

- Added a **Tree Graph** to roll-up appointments from **Doctor** → **Department** → **Hospital**, showing the aggregated count of appointments at each level.
- A **Card visualization** was used to display the **total count of appointments (120)**.

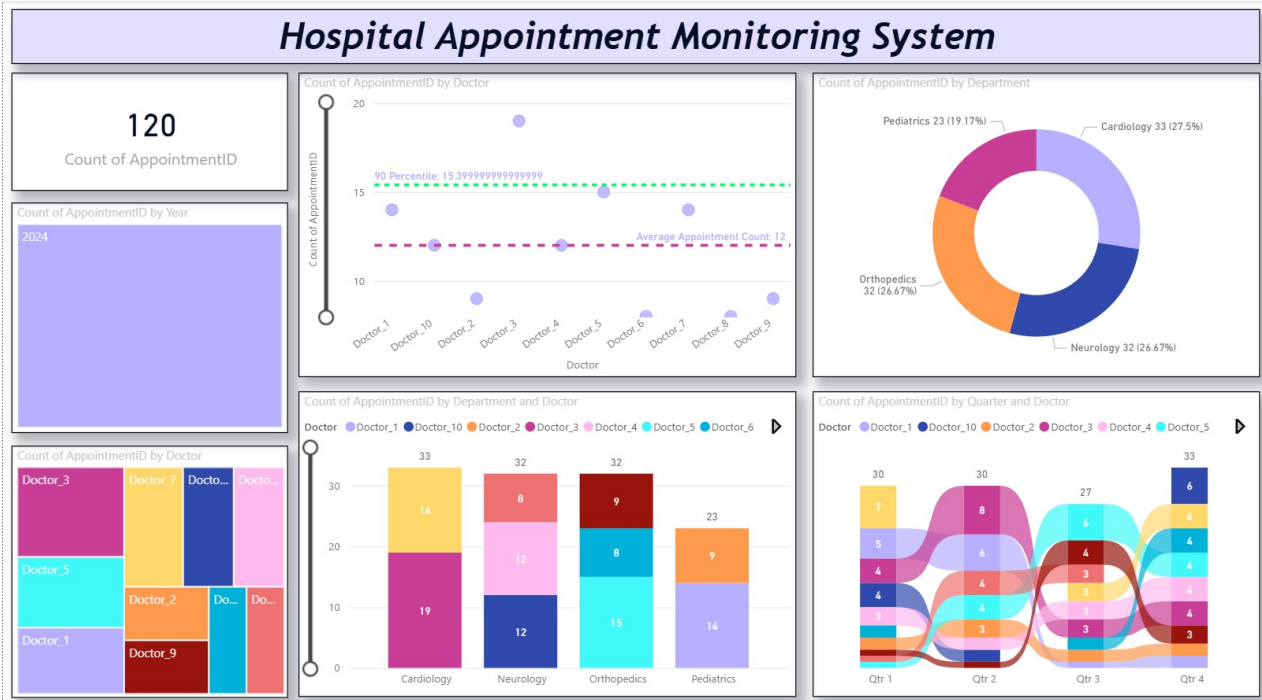


3. Additional Insights & Enhancements

Beyond the given requirements, additional visualizations and features were added for deeper insights:

- **Scatter Plot:**
 - Plots appointments per doctor.
 - Includes an **Average Appointment Count line** and a **90th Percentile line** to highlight performance benchmarks.
 - Added **Zoom Slider** for interactive filtering.
- **Bar Chart (Stacked by Department & Doctor):**
 - Displays count of appointments grouped by department and subdivided by doctors.
 - Appointment counts are shown as **values on the bars**.
 - Added **Zoom Slider** for detailed exploration.
- **Donut Chart:**
 - Visualizes the **percentage distribution of appointments** across departments.
- **Ribbon Chart:**
 - Shows the trend of appointments per doctor across quarters.

- Helps identify doctor-wise appointment dominance over time.
- **Treemap:**
 - Quick overview of appointments by doctor with proportional sizing.



4. Key Takeaways

- Cardiology and Neurology show the **highest number of appointments (33 & 32)**, while Paediatrics is comparatively lower.
- Doctor-wise distribution highlights workload imbalance, with certain doctors handling significantly more patients.
- The scatter plot lines (average & 90th percentile) help in identifying doctors performing above or below benchmarks.
- The dashboards enable **multi-level drill-downs**, **time-series tracking**, and **departmental comparisons** for hospital management.